On a finite-free Stam-type inequality for a root-interaction functional

Abstract

Let p and q be monic real-rooted polynomials of degree n. The *symmetric finite free additive convolution* $p \boxplus_n q$ is the monic degree-n polynomial whose coefficients are given by

$$(p \boxplus_n q)(x) = \sum_{k=0}^{n} c_k x^{n-k}, \qquad c_k = \sum_{i+j=k} \frac{(n-i)!(n-j)!}{n!(n-k)!} a_i b_j,$$

where $p(x) = \sum_{k=0}^{n} a_k x^{n-k}$ and $q(x) = \sum_{k=0}^{n} b_k x^{n-k}$. For a monic polynomial $p(x) = \prod_{i=1}^{n}(x - \lambda_i)$ with simple roots we consider

$$\Phi_n(p) := \sum_{i=1}^{n} \Big(\sum_{j \neq i} \frac{1}{\lambda_i - \lambda_j} \Big)^2,$$

and set $\Phi_n(p) = \infty$ when p has a multiple root. The question is whether the *finite-free Stam inequality*

$$\frac{1}{\Phi_n(p \boxplus_n q)} \geq \frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(q)}$$

holds for all monic real-rooted p, q of degree n.

We prove the inequality with equality for $n = 2$, we prove it for $n = 3$ by an explicit translation-invariant formula, and we prove that equality holds for all n on the (scaled) probabilists' Hermite family, which plays the role of the finite-free Gaussian. For general $n \geq 4$ we do not know a complete proof, but we record structural identities and numerical evidence.

1 The convolution $\boxplus_n$ and a linearizing transform

Fix $n \geq 1$. Let

$$p(x) = \sum_{k=0}^{n} a_k x^{n-k}, \qquad q(x) = \sum_{k=0}^{n} b_k x^{n-k}$$

be monic degree-n polynomials ($a_0 = b_0 = 1$). Their symmetric finite free additive convolution is

$$(p \boxplus_n q)(x) = \sum_{k=0}^{n} c_k x^{n-k}, \qquad c_k := \sum_{i+j=k} \frac{(n-i)!(n-j)!}{n!(n-k)!} a_i b_j.$$

This is the "symmetric additive convolution" of Marcus–Spielman–Srivastava [1] (up to sign conventions).

Remark 1 (A linearizing transform in a truncated polynomial ring). Define

$$\mathcal{T}_n(p)(t) := \sum_{k=0}^{n} \alpha_k t^k, \qquad \alpha_k := \frac{(n-k)!}{n!} a_k. \tag{1}$$

Then for $r = p \boxplus_n q$ one has

$$\mathcal{T}_n(r) \equiv \mathcal{T}_n(p)\,\mathcal{T}_n(q) \pmod{t^{n+1}}. \tag{2}$$

Equivalently, $\mathcal{T}_n$ is a ring isomorphism between $(\mathbb{C}[x]_{\le n}, \boxplus_n)$ and the truncated ring $(\mathbb{C}[t]/(t^{n+1}), \cdot)$. Indeed, writing $\alpha_k = (n-k)!a_k/n!$ and $\beta_k = (n-k)!b_k/n!$, the defining convolution formula gives for each $0 \le k \le n$,

$$\frac{(n-k)!}{n!}c_k = \sum_{i+j=k} \frac{(n-i)!}{n!}a_i \, \frac{(n-j)!}{n!}b_j,$$

which is exactly the Cauchy product rule for coefficients of the product $\mathcal{T}_n(p)\mathcal{T}_n(q)$, truncated to degrees $\le n$.

2 The functional Φ_n and elementary identities

Let $p(x) = \prod_{i=1}^n (x - \lambda_i)$ be monic with simple real roots $\lambda_1, \dots, \lambda_n$. Set

$$H_i := \sum_{j \ne i} \frac{1}{\lambda_i - \lambda_j}.$$

Then $\Phi_n(p) = \sum_{i=1}^n H_i^2$.

Lemma 1 (Derivative formula at roots)**.** *If p has simple roots, then for each i one has*

$$\frac{p''(\lambda_i)}{p'(\lambda_i)} = 2\sum_{j \ne i} \frac{1}{\lambda_i - \lambda_j} = 2H_i. \tag{3}$$

Consequently,

$$\Phi_n(p) = \frac{1}{4}\sum_{i=1}^n \Big(\frac{p''(\lambda_i)}{p'(\lambda_i)}\Big)^2. \tag{4}$$

Proof. Write $p(x) = (x - \lambda_i)g_i(x)$ where $g_i(x) = \prod_{j \ne i}(x - \lambda_j)$. Then $p'(x) = g_i(x) + (x - \lambda_i)g_i'(x)$ and $p''(x) = 2g_i'(x) + (x - \lambda_i)g_i''(x)$. Evaluating at $x = \lambda_i$ gives $p'(\lambda_i) = g_i(\lambda_i)$ and $p''(\lambda_i) = 2g_i'(\lambda_i)$. Finally, by logarithmic differentiation of g_i,

$$\frac{g_i'(\lambda_i)}{g_i(\lambda_i)} = \sum_{j \ne i} \frac{1}{\lambda_i - \lambda_j}.$$

This yields (3) and hence (4). □

Lemma 2 (Cancellation of three-body terms)**.** *For distinct indices i, j, k one has*

$$\frac{1}{(\lambda_i - \lambda_j)(\lambda_i - \lambda_k)} + \frac{1}{(\lambda_j - \lambda_i)(\lambda_j - \lambda_k)} + \frac{1}{(\lambda_k - \lambda_i)(\lambda_k - \lambda_j)} = 0. \tag{5}$$

As a consequence,

$$\Phi_n(p) = \sum_{i \ne j} \frac{1}{(\lambda_i - \lambda_j)^2} = 2\sum_{1 \le i < j \le n} \frac{1}{(\lambda_i - \lambda_j)^2}. \tag{6}$$

Proof. Multiply (5) by $(\lambda_i - \lambda_j)(\lambda_j - \lambda_k)(\lambda_k - \lambda_i)$ to obtain

$$-(\lambda_j - \lambda_k) + (\lambda_i - \lambda_k) - (\lambda_i - \lambda_j) = 0.$$

This proves (5).

For (6), expand

$$\Phi_n(p) = \sum_i \Big(\sum_{j\neq i} \frac{1}{\lambda_i - \lambda_j}\Big)^2 = \sum_i \sum_{j\neq i} \frac{1}{(\lambda_i - \lambda_j)^2} + \sum_i \sum_{\substack{j,k\neq i\\ j\neq k}} \frac{1}{(\lambda_i - \lambda_j)(\lambda_i - \lambda_k)}.$$

The first term is $\sum_{i\neq j} 1/(\lambda_i - \lambda_j)^2$. The second term can be grouped by unordered triples $\{i, j, k\}$ and vanishes by (5). This yields (6). □

Remark 2 (Scaling and translation). If $p_c(x) := p(x - c)$ then $\Phi_n(p_c) = \Phi_n(p)$. If $p_s(x) := s^n p(x/s)$ for $s \neq 0$ (so the roots scale by s), then $\Phi_n(p_s) = \Phi_n(p)/s^2$. Thus $1/\Phi_n$ is translation invariant and homogeneous of degree 2 under dilation.

3 The case $n = 2$: equality

Theorem 1. *Let p, q be monic real-rooted quadratics. Then*

$$\frac{1}{\Phi_2(p \boxplus_2 q)} = \frac{1}{\Phi_2(p)} + \frac{1}{\Phi_2(q)}.$$

In particular, the finite-free Stam inequality holds with equality for $n = 2$.

Proof. Write

$$p(x) = x^2 + a_1 x + a_2 = (x - \lambda_1)(x - \lambda_2), \qquad q(x) = x^2 + b_1 x + b_2 = (x - \mu_1)(x - \mu_2).$$

By Lemma 2,

$$\Phi_2(p) = 2\,\frac{1}{(\lambda_1 - \lambda_2)^2} = \frac{2}{\mathrm{Disc}(p)},$$

where $\mathrm{Disc}(p) = (\lambda_1 - \lambda_2)^2 = a_1^2 - 4a_2$ is the discriminant. Thus $1/\Phi_2(p) = \mathrm{Disc}(p)/2$, and similarly for q.

The convolution coefficients for $n = 2$ are

$$(p \boxplus_2 q)(x) = x^2 + (a_1 + b_1)x + \Big(a_2 + b_2 + \frac{1}{2}a_1 b_1\Big).$$

Hence

$$\begin{aligned}\mathrm{Disc}(p \boxplus_2 q) &= (a_1 + b_1)^2 - 4\Big(a_2 + b_2 + \frac{1}{2}a_1 b_1\Big)\\ &= (a_1^2 - 4a_2) + (b_1^2 - 4b_2) = \mathrm{Disc}(p) + \mathrm{Disc}(q).\end{aligned}$$

Therefore

$$\frac{1}{\Phi_2(p \boxplus_2 q)} = \frac{\mathrm{Disc}(p \boxplus_2 q)}{2} = \frac{\mathrm{Disc}(p)}{2} + \frac{\mathrm{Disc}(q)}{2} = \frac{1}{\Phi_2(p)} + \frac{1}{\Phi_2(q)}.$$

□

4 The case $n = 3$: an explicit invariant formula and the inequality

For cubics it is convenient to use the translation-invariant depressed form.

Lemma 3 (Depressed parameters and additivity)**.** *Let*

$$p(x) = x^3 + a_1x^2 + a_2x + a_3.$$

Define the translation-invariant quantities

$$P := a_2 - \frac{a_1^2}{3}, \qquad Q := a_3 - \frac{a_1a_2}{3} + \frac{2a_1^3}{27}. \tag{7}$$

Then $p(x + a_1/3) = x^3 + Px + Q$*. Moreover, if* $r = p \boxplus_3 q$ *then* $P_r = P_p + P_q$ *and* $Q_r = Q_p + Q_q$*.*

Proof. The first statement is the classical Tschirnhaus shift. For additivity under $\boxplus_3$, compute from the defining coefficient formula

$$c_1 = a_1 + b_1, \qquad c_2 = a_2 + b_2 + \frac{2}{3}a_1b_1, \qquad c_3 = a_3 + b_3 + \frac{1}{3}(a_1b_2 + a_2b_1).$$

Substituting these into (7) for r yields $P_r = P_p + P_q$ and $Q_r = Q_p + Q_q$. □

Proposition 1 (A closed form for Φ_3)**.** *Let* $p(x) = x^3 + Px + Q$ *be a monic real-rooted cubic with distinct roots. Then*

$$\Phi_3(p) = \frac{18P^2}{\mathrm{Disc}(p)}, \qquad \textit{equivalently} \qquad \frac{1}{\Phi_3(p)} = \frac{\mathrm{Disc}(p)}{18P^2}, \tag{8}$$

where $\mathrm{Disc}(p) = -4P^3 - 27Q^2 > 0$*.*

Proof. Let the roots be $\lambda_1, \lambda_2, \lambda_3$. For the depressed cubic one has the elementary relations

$$\lambda_1 + \lambda_2 + \lambda_3 = 0, \qquad \lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3 = P.$$

By Lemma 2,

$$\Phi_3(p) = 2\sum_{i<j} \frac{1}{(\lambda_i - \lambda_j)^2}.$$

Set $\Delta_{ij} := \lambda_i - \lambda_j$. Then

$$\mathrm{Disc}(p) = (\Delta_{12}\Delta_{13}\Delta_{23})^2 \qquad \text{and} \qquad \sum_{i<j} \frac{1}{\Delta_{ij}^2} = \frac{\Delta_{12}^2\Delta_{13}^2 + \Delta_{12}^2\Delta_{23}^2 + \Delta_{13}^2\Delta_{23}^2}{(\Delta_{12}\Delta_{13}\Delta_{23})^2}.$$

The numerator

$$E := \Delta_{12}^2\Delta_{13}^2 + \Delta_{12}^2\Delta_{23}^2 + \Delta_{13}^2\Delta_{23}^2$$

is a symmetric homogeneous polynomial of degree 4 in $(\lambda_1, \lambda_2, \lambda_3)$. By the fundamental theorem of symmetric polynomials, any symmetric homogeneous polynomial of degree 4 in three variables is a polynomial in the elementary symmetric functions $\sigma_1, \sigma_2, \sigma_3$, and hence can be written as

$$A\sigma_1^4 + B\sigma_1^2\sigma_2 + C\sigma_2^2 + D\sigma_1\sigma_3.$$

Here $\sigma_1 = \lambda_1 + \lambda_2 + \lambda_3 = 0$, so $E = C\sigma_2^2$. Since $\sigma_2 = P$, we have $E = CP^2$. To determine C, evaluate at the triple $(\lambda_1, \lambda_2, \lambda_3) = (-1, 0, 1)$ (for which $P = -1$): one finds $\Delta_{12}^2 = 1$, $\Delta_{13}^2 = 4$, $\Delta_{23}^2 = 1$, hence $E = 4 + 1 + 4 = 9$. Thus $C = 9$ and therefore $E = 9P^2$.

Consequently,

$$\sum_{i<j} \frac{1}{\Delta_{ij}^2} = \frac{9P^2}{\mathrm{Disc}(p)} \qquad \Rightarrow \qquad \Phi_3(p) = 2 \cdot \frac{9P^2}{\mathrm{Disc}(p)} = \frac{18P^2}{\mathrm{Disc}(p)},$$

which is (8). □

Theorem 2 (Stam inequality for $n = 3$)**.** *Let p, q be monic real-rooted cubics. Then*

$$\frac{1}{\Phi_3(p \boxplus_3 q)} \geq \frac{1}{\Phi_3(p)} + \frac{1}{\Phi_3(q)}.$$

Proof. If p or q has a multiple root then $\Phi_3 = \infty$ and the right-hand side is unchanged while the left-hand side is nonnegative, so the inequality is trivial. Hence assume distinct roots.

By Lemma 3 we may translate so that $p(x) = x^3 + P_p x + Q_p$ and $q(x) = x^3 + P_q x + Q_q$. Set $r = p \boxplus_3 q$, so $r(x) = x^3 + P_r x + Q_r$ with $P_r = P_p + P_q$ and $Q_r = Q_p + Q_q$. By Proposition 1,

$$\frac{1}{\Phi_3(p)} = \frac{-4P_p^3 - 27Q_p^2}{18P_p^2} = -\frac{2}{9}P_p - \frac{3}{2}\frac{Q_p^2}{P_p^2},$$

and similarly for q and r. The linear P-terms are additive and cancel from the desired inequality. Thus it suffices to prove

$$\frac{(Q_p + Q_q)^2}{(P_p + P_q)^2} \leq \frac{Q_p^2}{P_p^2} + \frac{Q_q^2}{P_q^2}. \tag{9}$$

For real-rooted depressed cubics with distinct roots one has $P < 0$ (indeed $P = \sum_{i<j} \lambda_i \lambda_j = -\frac{1}{2}\sum_i \lambda_i^2 < 0$). Hence write $P_p = -\alpha$ and $P_q = -\beta$ with $\alpha, \beta > 0$. Then (9) becomes

$$\frac{(Q_p + Q_q)^2}{(\alpha + \beta)^2} \leq \frac{Q_p^2}{\alpha^2} + \frac{Q_q^2}{\beta^2}.$$

Set $x := Q_p/\alpha$ and $y := Q_q/\beta$. Then $Q_p + Q_q = \alpha x + \beta y$, and the desired inequality reads

$$\frac{(\alpha x + \beta y)^2}{(\alpha + \beta)^2} \leq x^2 + y^2. \tag{10}$$

By Cauchy–Schwarz,

$$(\alpha x + \beta y)^2 \leq (\alpha^2 + \beta^2)(x^2 + y^2) \leq (\alpha + \beta)^2(x^2 + y^2),$$

where the last inequality uses $\alpha^2 + \beta^2 \leq (\alpha + \beta)^2$. Dividing by $(\alpha + \beta)^2$ gives (10), completing the proof. □

5 Equality for the Hermite family (all n)

Let He_n denote the probabilists' Hermite polynomial. For $\sigma > 0$ define the scaling

$$\mathrm{He}_{n,\sigma}(x) := \sigma^n \mathrm{He}_n(x/\sigma).$$

Its roots are real and simple.

Proposition 2 (Hermite is stable under $\boxplus_n$)**.** *For all $\sigma, \tau > 0$ one has*

$$\mathrm{He}_{n,\sigma} \boxplus_n \mathrm{He}_{n,\tau} = \mathrm{He}_{n,\sqrt{\sigma^2+\tau^2}}.$$

Proof. We compute the $\mathcal{T}_n$-transform (1). The probabilists' Hermite polynomial admits the explicit expansion

$$\mathrm{He}_n(x) = n! \sum_{m=0}^{\lfloor n/2 \rfloor} \frac{(-1)^m}{m!(n-2m)!2^m} x^{n-2m}.$$

Therefore

$$\mathrm{He}_{n,\sigma}(x) = \sigma^n \mathrm{He}_n(x/\sigma) = n! \sum_{m=0}^{\lfloor n/2 \rfloor} \frac{(-1)^m \sigma^{2m}}{m!(n-2m)!2^m} x^{n-2m}.$$

Thus the coefficients a_k vanish for odd k, and for $k = 2m$ we have

$$a_{2m} = n! \frac{(-1)^m \sigma^{2m}}{m!(n-2m)!2^m}.$$

Hence

$$\alpha_{2m} = \frac{(n-2m)!}{n!} a_{2m} = \frac{(-1)^m \sigma^{2m}}{m! 2^m}, \qquad \alpha_{2m+1} = 0.$$

So, in the truncated ring $\mathbb{C}[t]/(t^{n+1})$,

$$\mathcal{T}_n(\mathrm{He}_{n,\sigma})(t) = \sum_{m=0}^{\lfloor n/2 \rfloor} \frac{(-1)^m}{m!} \Big(\frac{\sigma^2 t^2}{2}\Big)^m \equiv \exp\Big(-\frac{\sigma^2 t^2}{2}\Big) \pmod{t^{n+1}}.$$

Therefore

$$\mathcal{T}_n(\mathrm{He}_{n,\sigma} \boxplus_n \mathrm{He}_{n,\tau}) \equiv \mathcal{T}_n(\mathrm{He}_{n,\sigma})\, \mathcal{T}_n(\mathrm{He}_{n,\tau}) \equiv \exp\Big(-\frac{(\sigma^2+\tau^2)t^2}{2}\Big) \equiv \mathcal{T}_n(\mathrm{He}_{n,\sqrt{\sigma^2+\tau^2}}) \pmod{t^{n+1}}.$$

Since $\mathcal{T}_n$ is injective (one recovers a_k from α_k by $a_k = \frac{n!}{(n-k)!}\alpha_k$), the claim follows. □

Proposition 3 (Exact computation of Φ_n for Hermite)**.** *For $\sigma > 0$,*

$$\Phi_n(\mathrm{He}_{n,\sigma}) = \frac{n(n-1)}{4\sigma^2}, \qquad \textit{hence} \qquad \frac{1}{\Phi_n(\mathrm{He}_{n,\sigma})} = \frac{4\sigma^2}{n(n-1)}.$$

Proof. Let $h_1, \ldots, h_n$ be the roots of He_n. The differential equation for He_n is

$$\mathrm{He}_n''(x) - x\mathrm{He}_n'(x) + n\mathrm{He}_n(x) = 0.$$

Evaluating at $x = h_i$ (where $\mathrm{He}_n(h_i) = 0$) gives $\mathrm{He}_n''(h_i) = h_i \mathrm{He}_n'(h_i)$. By Lemma 1,

$$2\sum_{j \neq i} \frac{1}{h_i - h_j} = \frac{\mathrm{He}_n''(h_i)}{\mathrm{He}_n'(h_i)} = h_i, \qquad \text{i.e.} \qquad \sum_{j \neq i} \frac{1}{h_i - h_j} = \frac{h_i}{2}.$$

Therefore

$$\Phi_n(\mathrm{He}_n) = \sum_{i=1}^n \Big(\frac{h_i}{2}\Big)^2 = \frac{1}{4} \sum_{i=1}^n h_i^2.$$

On the other hand, the coefficient of x^{n-2} in He_n is $-\binom{n}{2}$ (visible from the explicit series above). Since He_n is monic and has $\sum_i h_i = 0$, we have

$$\sum_{i<j} h_i h_j = -\frac{1}{2}\sum_i h_i^2,$$

and also $\sum_{i<j} h_i h_j$ equals the coefficient of x^{n-2}. Thus $-\frac{1}{2}\sum_i h_i^2 = -\binom{n}{2}$, i.e. $\sum_i h_i^2 = n(n-1)$. Hence $\Phi_n(\mathrm{He}_n) = n(n-1)/4$.

For $\mathrm{He}_{n,\sigma}(x) = \sigma^n \mathrm{He}_n(x/\sigma)$, the roots scale by σ, so by the scaling remark after Lemma 2 we get $\Phi_n(\mathrm{He}_{n,\sigma}) = \Phi_n(\mathrm{He}_n)/\sigma^2 = n(n-1)/(4\sigma^2)$. □

Corollary 1 (Equality along Hermite)**.** *For all $\sigma, \tau > 0$,*

$$\frac{1}{\Phi_n(\mathrm{He}_{n,\sigma} \boxplus_n \mathrm{He}_{n,\tau})} = \frac{1}{\Phi_n(\mathrm{He}_{n,\sigma})} + \frac{1}{\Phi_n(\mathrm{He}_{n,\tau})}.$$

Proof. Combine Proposition 2 and Proposition 3:

$$\frac{1}{\Phi_n(\mathrm{He}_{n,\sigma} \boxplus_n \mathrm{He}_{n,\tau})} = \frac{4(\sigma^2+\tau^2)}{n(n-1)} = \frac{4\sigma^2}{n(n-1)} + \frac{4\tau^2}{n(n-1)}.$$

□

6 Status for general n

Remark 3 (What is proved here). The preceding results establish the finite-free Stam inequality for $n = 2$ and $n = 3$, and show that equality holds for all n on the Hermite family.

Remark 4 (Numerical evidence). A brute-force numerical search over random real-rooted polynomials (generated by random distinct real roots) for degrees up to $n = 10$ did not produce a counterexample to

$$\frac{1}{\Phi_n(p \boxplus_n q)} \geq \frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(q)}.$$

While this is not a proof, it is consistent with the conjecture.

Remark 5 (Open problem). For general $n \geq 4$, the inequality remains open to the author. The low-degree proofs reduce the claim to explicit inequalities in finitely many translation-invariant parameters which are additive under $\boxplus_n$. For $n \geq 4$ there are more such invariants (finite-free cumulants), and it is unclear how to express Φ_n in terms of them in a way suitable for a general inequality.

References

[1] A. W. Marcus, D. A. Spielman, and N. Srivastava, *Finite free convolutions of polynomials*, Probab. Theory Related Fields **182** (2022), no. 3–4, 807–848.

[2] A. W. Marcus, *Polynomial convolutions and (finite) free probability*, preprint (available from the author's webpage).